



BRyA LUNA, APAYAO October 9-10 **2025** **Research Summit**

SUMMIT PROCEEDINGS

Toward a Knowledge-Based Future: Research
Review and Agenda-Setting for the UNESCO
Biosphere Reserve of the yApayaos



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PART I: OVERVIEW AND CONTEXT

Summit Background, Rationale, and Objectives

The Biosphere Reserve of the yApayaos (BRyA) was designated under **UNESCO's Man and the Biosphere (MAB) Programme on July 5, 2024**. This marks the province as a globally recognized site for integrating conservation, sustainable development, and cultural resilience. Thus, governance and management of the reserve must be informed by a robust, interdisciplinary, and Indigenous knowledge-grounded research agenda.

To ensure alignment of academic contributions with the reserve's needs, the BRyA Research Summit entitled “**Toward a Knowledge-Based Future: Research Review and Agenda-Setting for the UNESCO Biosphere Reserve of the yApayaos**” was developed to review recent research outputs from higher education institutions and co-create a forward-looking research agenda responsive to the biocultural context and governance priorities of the biosphere reserve.

The research summit was developed in alignment with the following objectives:

1. To synthesize academic research conducted in Apayao over the last five years (2020–2025) across themes such as biodiversity, Indigenous knowledge, climate resilience, livelihoods, land use, and natural resource management.
2. To assess the alignment of existing research with the core functions of a UNESCO Biosphere Reserve: conservation, development, and logistic support.
3. To co-develop a five-year research agenda (2026–2031) addressing gaps and priority knowledge needs for the governance and adaptive management of the Apayao Biosphere Reserve.



The 2025 BRyA Research Summit provided a strategic opportunity to anchor the governance of the Biosphere Reserve of the yApayaos in a sound and inclusive knowledge base. By drawing from existing academic work and projecting future needs, we ensure that the reserve fulfills its mandate as a living laboratory for sustainability, rooted in both science and culture.

The findings from the discussions and presentations to be elaborated further in this document will aid in the development of the following key outputs for the planning and management of the biosphere reserve:

1. **Consolidated Research Review Report:** An annotated bibliography and thematic synthesis of academic research on Apayao within the last 10 years from higher education institutions (HEIs), research organizations, and government agencies
2. **Five-Year Research Agenda (2026–2031):** A document outlining priority topics, guiding questions aligned with the governance and management needs of the BRyA, target partnerships, proposed methods, and knowledge mobilization pathways
3. **Conference Proceedings and Policy Brief:** A document providing advocacy materials and a summary of recommendations for research funding, policy support, and stakeholder collaboration





PART II: SUMMIT PROPER

Plenary Presentations

Summary of Plenary Research Presentations

Bringing together academic institutions, government agencies, and civil society partners, the plenary research presentations showcased a diverse range of studies, including natural resource utilization, climate impact assessments, ecological surveys, cultural practices, and policy frameworks. The presentations can be categorized into five themes that can be found listed below.

Sustainable Resource Use and Climate Research

Student researchers from Apayao State College presented studies highlighting sustainable management of local resources and climate resilience. Topics included the utilization of Anabo fibers as a non-timber forest product, tapping forest food crops and indigenous food resources for food security, and applying natural dye-yielding plants in Luna. Several studies also focused on the impacts of extreme weather events, coping mechanisms of farmers and fisherfolk, agroforestry crop diversity, and indicators of sustainable development in community-based forest management programs.

Biodiversity, Ecology, and Traditional Knowledge

The Apayao State College also presented studies on biodiversity and ecological dynamics of Apayao, including studies on fern diversity, Almaciga seedling survival, ethnomedicinal plant properties, and traditional Isneg river conservation practices (Sannad). Ateneo de Manila University complemented these efforts with their work on freshwater biodiversity and aquatic insect studies, reporting 15 new species across three insect orders within the province of Apayao. The Department of Environment and Natural Resources – Watershed and Water Resources Research Development and Extension Center (DENR-WWRDEC) contributed





documentation of Indigenous weather and climate forecasting knowledge from the Isneg tribe.

Policy, Planning, and Development Frameworks

Presentations from DENR-CAR and the Department of Economy, Planning, and Development (DEPDev) focused on post-disaster recovery and regional sustainability planning. These included the Green Assessment & Recovery Framework Pilot Study after Typhoon “Ompong,” the CAR Regional Physical Framework Plan 2024–2054, and the Cordillera Regional Development Plan 2023–2028, along with the Regional Research, Development, and Innovation Agenda.

Complementary Sectoral Research

Other agencies and institutions presented additional research reinforcing the aforementioned themes. DepEd-Apayao shared a floral inventory of marsh ecosystems, while DOST-Apayao introduced projects on e-transportation, sustainable agriculture, and other projects within the province. The National Research Council of the Philippines (NRCP) under DOST highlighted research on microalgae as a biofuel feedstock alongside a presentation on NRCP membership. The Mariano Marcos State University (MMSU) discussed studies focused on ethnobotany in the region while the National Museum of the Philippines shared existing data on biodiversity within the province.

Species Rediscoveries and Conservation Milestones

The Philippine Eagle Foundation (PEF) also presented key biodiversity findings in Apayao from 2020–2025, including updates on the population of the Philippine Eagle within the province and recent milestones on biodiversity research such as the rediscovery of *Pseudapocrytus schadenbergi* (Heller, 1912) after 111 years. The presentation underscores Apayao’s importance as a biodiversity hotspot and a critical area for conservation research.





Breakout Room Sessions

Biodiversity Research and Conservation

Facilitator: Dr. Jhoana M. Garces (Ateneo de Manila University)

Documenter: Raphael Deza, Andie Penafior

- Participants: Liezel M. Magtoto, Pierangeli Villan, Paul Alfonso Onda, Clarissa Mae Escalona, Mary Chalette Fresco, Marivene Santos, Treeland Columna, Ralph Verzon, Tosaphol Saetung Keetapitchayakul, Dr. Jayson Ibañez, Angela Castillo, Grace de los Reyes, Loida de Guzman, Sarah Mae Batu, Andrea Joyce Peñaflor, Erika Jane Acebes, Breiden Dela Cruz, Arnold Janssen Castillo, Jamaica Q. Ramos, Emmanuel John Estelon, Ma. Rocielle Libnao, Alice Pagaling, Ria Balbin, Naville Tacbadal, Marimar Rodriguez, Yra Pagala, Carlo Dave Yutoc, Paulo Nicolas, Marcel Tomas, Dennis Dizon, Lorenzo Deza, Jerzon Sebastian, Debbie Glynn Manangkil (DOST), and other representatives from academe, LGUs, and civil society.

Session Overview

This breakout room session aimed to identify the key strengths and weaknesses and translate these insights into priority areas for future research and project development. Participants during the session were given notecards to write down insights on research strengths and gaps with the goals of identifying underrepresented taxa, aligning research initiatives, and defining priority areas for the next five to ten years.

Policy Recommendations for the Apayao Biosphere Reserve

1. Conduct of Baseline Biodiversity Studies
 - a. Promotion of baseline studies for all taxa groups, but especially for the groups listed below that are greatly underrepresented. Linkages and higher level interactions such as invasive species and symbiosis alongside studies on optimizing and testing monitoring systems for biodiversity studies were also recommended by the breakout session group.





- i. microbiology
- ii. ethnobotany
- iii. bryophytes, lichens, and plants (in general)
- iv. invertebrates
- v. reptiles
- vi. volant and non-volant mammals
- vii. updates on bird inventories (including migratory birds)
- viii. aquatic vertebrates (e.g. ichthyofauna)

2. Community Inclusion and Co-authorship Policy

- a. Formal recognition for IP leaders and local field experts, either as co-authors or through acknowledgment, depending on their contribution.
- b. Nomination of local expert in community to be trained and compensated
- c. Integration of biodiversity education into local curricula through iNaturalist and youth-led documentation initiatives.

3. Institutionalization of the BRyA Research Center

- a. Envisioned as a permanent hub for biodiversity research, capacity building, and hosting external researchers to increase collaboration and capacity building with local community

4. Creation of a Centralized Biodiversity Database/Website

- a. Linked to the provincial LGU website, featuring a science communication officer and incorporating NDAs and copyright protocols to protect sensitive data
- b. Creation of list of research encompassing different fields
- c. Creation of Job Item: Science Communication Officer in Biodiversity Research Center

5. Streamlining of Forms and Documentation

- a. Implement a regulatory function for the BRyA Council, including the release of clearance to all projects
 - i. Expedited review process for priority research areas (i.e. pre-approved status)





6. Formation of a BRyA Research Consortium

- a. A network of researchers, universities, and LGUs to sustain collaboration and convene future summits.

7. Conservation Incentives for Private Landowners

- a. Property tax exemptions and recognition of "conservation estates."
- b. Engagement of private landowners in negotiating and mediating land use issues





Land Use and Natural Resource Management

Facilitator: Ms. Hannie Taylan (Apayao State College)

Documenter: Randy Begtang, Bea

- Participants: Representatives from Apayao State College, UP Baguio, and the Provincial Government of Apayao

Session Overview

The breakout session aimed to identify challenges in land governance, policy harmonization, and sustainable land utilization. This breakout session identified four key areas and discussion points: Governance and Policy Coordination, Natural Resources and Biodiversity Management, Suitability of Land Uses, and Community Education and Public Awareness (CEPA).

Governance and Policy Coordination

Focuses on enhancing institutional coordination, policy coherence, participatory decision-making through evidence-based studies that strengthen transparent, accountable and inclusive governance for sustainable resource development and resource management.

Key Issues and Challenges

1. Issues in Zoning
 - a. Multi-issuance of tenurial instruments
 - b. Land area discrepancies due to DBM allocation
 - c. BRyA Core Zones do not reflect recent municipal boundary surveys
 - d. Harmonization of BRyA and Land Use Zones
2. Land Disputes
 - a. NCIP Conflicts
 - b. Kaingin Land Use Conversion





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Research Agenda

1. Formal documentation of governmental and policy disputes
2. Review of overlapping policies and authorizations between departments/agencies
3. Harmonization of existing and new policies, especially on land use zonation of BRyA

Natural Resources and Biodiversity Management

Focuses on promoting sustainable use, conservation, and restoration of ecosystems through studies on forest and watershed management, biodiversity protection, climate resilience, policy enhancement, and community participation to ensure ecological balance and long-term resource sustainability.

Key Issues and Challenges

1. Unsustainable Resource Management
 - a. Forest extraction, mining, quarrying
 - b. Unsustainable logging
2. Lack of knowledge about ecosystem services
 - a. Absence of payment for ecosystem services
 - b. Lack of wood requirements for each municipality
 - c. Lack of quantitative knowledge

Research Agenda

1. Assessment of land use implications to biodiversity conservation and sustainable management of natural resources
2. Conduct community-based valuation of ecosystem services

Suitability of Land Uses

Focuses on assessing and optimizing the use of land resources through scientific studies on its biological, socioeconomic, and environmental characteristics to promote sustainable and balanced resource management. It aims to guide evidence-based planning and policy formulation for optimal land utilization, development, and management decisions.





Key Issues and Challenges

1. Lack of Institutionalized Assessment Body
 - a. No agency assigned to provide suitability assessments
2. Lack of Research and Studies
 - a. Need for studies on land use suitability for development and agricultural purposes
 - b. Necessity for studies on agricultural use change due to climate change
3. Limited A&D Zones

Research Agenda

1. Investigation of land use suitability for development, agricultural, and disaster mitigation purposes

Community Education and Public Awareness (CEPA)

Focuses on strengthening public understanding, engagement, and collective action in environmental protection and resource management through studies on knowledge, attitudes, practices, and community-based approaches that promote inclusive and sustainable development.

Key Issues and Challenges

1. Lack of Monitoring and Evaluation of Policies
 - a. Effectiveness of policies is not known
 - b. Awareness of stakeholders is not measured
2. Limited Awareness and Participation
 - a. Public participation is limited especially without incentives

Research Agenda

1. Identification of strategies to evoke public participation in land use management
2. Evaluation of stakeholder awareness and response to land use policies





Plenary Presentations of Breakout Room Outputs

The plenary session of the BRyA Research Summit served as a venue for each breakout group to present the consolidated results of their thematic discussions. Facilitators and rapporteurs summarized their group outputs through presentations, highlighting major findings, research gaps, and recommended actions for inclusion in the BRyA 5-Year Research Agenda.

Presenters:

- Dr. Jhoana M. Garces – Biodiversity Research and Conservation
- Ms. Hannie Taylan – Land Use and Natural Resource Management
- Mr. Benjamin D. Vega – Climate Change and Ecosystem Resilience
- Dr. Zenaida Baoanan – Sociocultural Systems and IKSPs

Synthesis of Recommendations

The following synthesis presents the common recommendations that emerged from the documentation and analysis of all four breakout sessions. These themes were identified after the summit proceedings, during the consolidation of group reports and minutes by the BRyA Secretariat and partner institutions. They represent shared priorities and cross-cutting directions for the formulation of the Apayao BRyA Five-Year Research and Development Agenda (2025–2030).

Table N. Synthesis of Key Recommendations from Breakout Room Sessions

Common Theme	Key Recommendations	Lead Collaborators
Institutional Strengthening and Research Coordination	1. Establish the BRyA Research Center as a coordinating hub for research and capacity-building 2. Form a BRyA Research Consortium among LGUs, academe, and national agencies 3. Streamline research clearance and permitting under the BRyA Management Council with FPIC compliance	BRyA Council, LGUs, ASC, ADMU, UPB, DENR, NCIP



Integrated and Accessible Data Systems	1. Develop a centralized digital database for biodiversity, climate, land-use, and IKSP information 2. Implement data-sharing protocols, NDAs, and citation guidelines 3. Collaborate with national agencies for data integration and validation	BRyA Research Center, DOST, DENR, NCIP, NEDA, PAGASA, HEIs
Community Empowerment and Indigenous Participation	1. Ensure co-authorship and recognition for IPs and local collaborators 2. Build capacity of IP youth and local researchers through training and mentorships 3. Institutionalize community participation in research planning and implementation	NCIP, LGUs, Isnag Council of Elders, DepEd, CHED, NGOs
Policy Harmonization and Governance	1. Align sectoral policies (agriculture, forestry, land use, biodiversity) with BRyA priorities 2. Harmonize local zoning ordinances with BRyA's core, buffer, and transition zones 3. Integrate Lapat System and IKSPs into local environmental governance	PLGU Apayao, DENR, DA, NCIP, DILG, Municipal LGUs
Communication, Education,, and Public Awareness (CEPA)	1. Integrate BRyA content into DepEd and CHED curricula 2. Produce child-friendly IEC materials and campaigns on biodiversity and climate action. 3. Encourage youth engagement through eco-clubs, citizen science, and school partnerships	DepEd, CHED, ASC, ADMU, ASHS, LYDOs, PEF
Sustainable Livelihoods and Climate Resilience	1. Promote biodiversity-friendly enterprises (eco-tourism, agroforestry, green crafts) 2. Mainstream climate-smart agriculture and watershed rehabilitation 3. Expand renewable energy projects (hydro, solar, waste-to-energy)	LGUs, MENROs, DENR, DA, DTI, NEDA, NGOs, PGO



Monitoring, Evaluation, and Continuing Collaboration	1. Develop clear performance indicators and annual progress reviews 2. Hold annual BRyA Research Summits to share updates and results. 3. Document and publish BRyA Research Reports for transparency and learning	BRyA Secretariat, LGUs, Academe, Partner NGOs
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